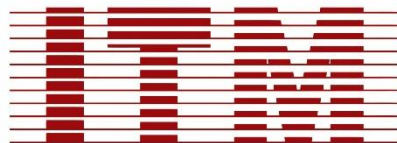




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गुण पुरस्कृत

DEPARTMENT OF ELECTRONICS & COMM. ENGINEERING
SUBJECT NAME- Sensors and Actuator Devices for IoT Lab
SUBJECT CODE- IO605



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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Subject: IO605 Sensors and Actuator Devices for IoT Lab

Name : XXXXXX XXXXXXXXXXXXX
Roll No. : 0905IO2310XX
Course & Sem : B.Tech. VI Semester
Branch : Computer Science & IOT

Submitted to:

Mr. Rajkumar Rajoria

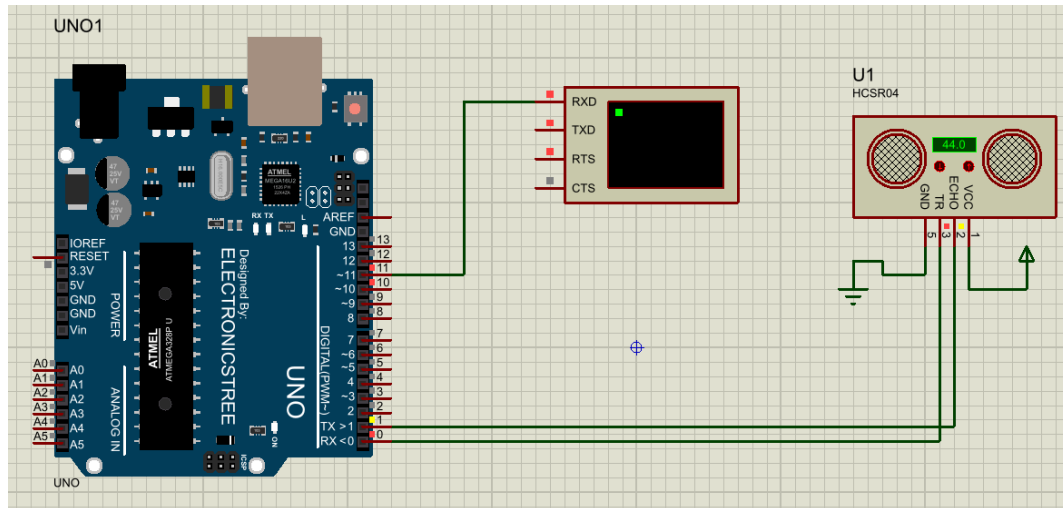
Asst. Professor, Dept. of ECE

Experiment - 1

Objective:

To measure the distance of an object using an ultrasonic sensor by transmitting and receiving sound waves, and to analyze the time-of-flight principle.

Circuit -



Code -

```
const int trigPin = 9;
const int echoPin = 10;
long duration;
float distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.begin(9600); // Initialize serial communication
}

void loop() {
  // Clear the trigPin
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  // Set the trigPin on HIGH state for 10 micro seconds
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
```



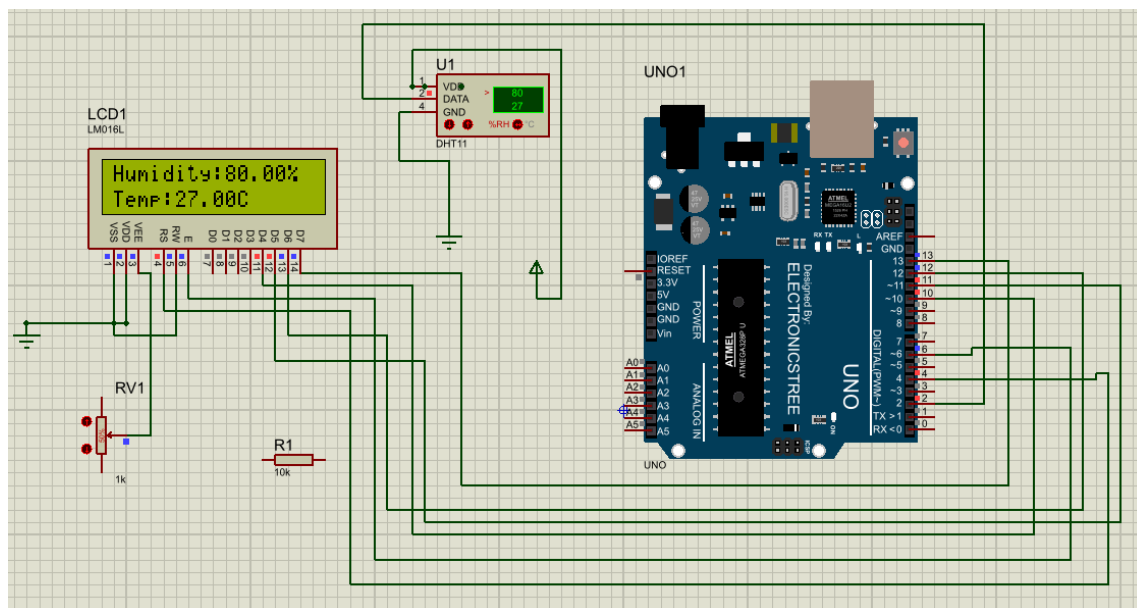
```
digitalWrite(trigPin, LOW);  
// Read the echoPin, returns the sound wave travel time in  
microseconds  
duration = pulseIn(echoPin, HIGH);  
// Calculating the distance (speed of sound is 343 m/s)  
distance = duration * 0.0343 / 2;  
// Print distance to Serial Monitor  
Serial.print("Distance: ");  
Serial.print(distance);  
Serial.println(" cm");  
  
delay(500); // Small delay to avoid interference
```

Experiment - 2

Objective:

To interface a temperature and humidity sensor and measure environmental conditions, and to understand digital sensing and calibration techniques.

Circuit -



Code -

```
#include <DHT.h>
#include <LiquidCrystal.h>

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

LiquidCrystal lcd(4, 6, 10, 11, 12, 13);

void setup() {
    lcd.begin(16, 2);
    lcd.clear();
    dht.begin();
```

```
}
```

```
void loop() {
```

```
    float h = dht.readHumidity();
```

```
    float t = dht.readTemperature();
```

```
    lcd.setCursor(0,0);
```

```
    lcd.print("Humidity:");
```

```
    lcd.print(h);
```

```
    lcd.print("%");
```

```
    lcd.setCursor(0,1);
```

```
    lcd.print("Temp:");
```

```
    lcd.print(t);
```

```
    lcd.print("C");
```

```
    delay(1000);
```

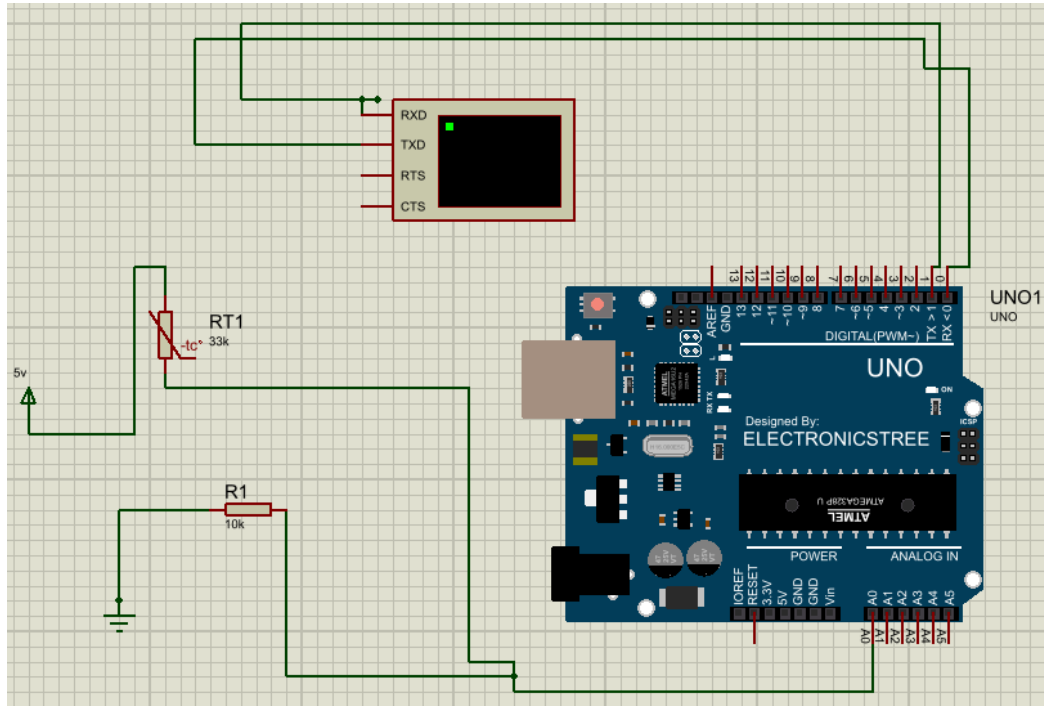
```
}
```

Experiment - 3

Objective:

To test the working of a thermistor and measure temperature variations by analyzing resistance changes with respect to temperature.

Circuit -



Code -

```
#include <math.h>

const int thermistorPin = A0;
const float R_fixed = 10000.0; // Fixed resistor 10k
const float V_in = 5.0;

void setup() {
    Serial.begin(9600);
}

void loop() {
    int adcVal = analogRead(thermistorPin);

    // Calculate thermistor resistance
    float V_out = (adcVal * V_in) / 1023.0;
```



```
float R_th = R_fixed * (V_in / V_out - 1.0);

// Steinhart-Hart Equation to get temperature in Celsius
float logR = log(R_th);
float T = 1.0 / (0.001129148 + (0.000234125 * logR) + (0.0000000876741
* logR * logR * logR));
T = T - 273.15; // Convert Kelvin to Celsius

Serial.print("Temperature: ");
Serial.print(T);
Serial.println(" C");

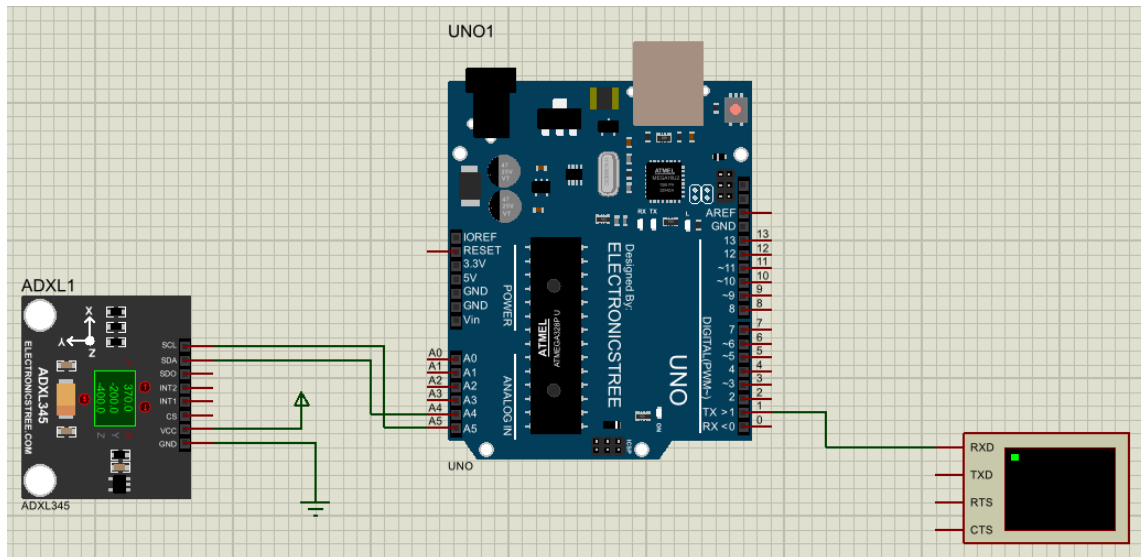
delay(1000);
}
```

Experiment - 4

Objective:

To interface the ADXL345 accelerometer and measure acceleration in three axes, and to understand MEMS sensor operation and motion detection.

Circuit -



Code -

```
#include <Wire.h>

#include <Adafruit_Sensor.h>
#include <Adafruit_ADXL345_U.h>

Adafruit_ADXL345_Unified accel1 = Adafruit_ADXL345_Unified(1); // LOW (0x53)
Adafruit_ADXL345_Unified accel2 = Adafruit_ADXL345_Unified(2); // HIGH (0x1D)

void setup() {
  Serial.begin(9600);
  if (!accel1.begin(0x53)) Serial.println("Sensor 1 not found!");
  if (!accel2.begin(0x1D)) Serial.println("Sensor 2 not found!");
}

void loop() {
```



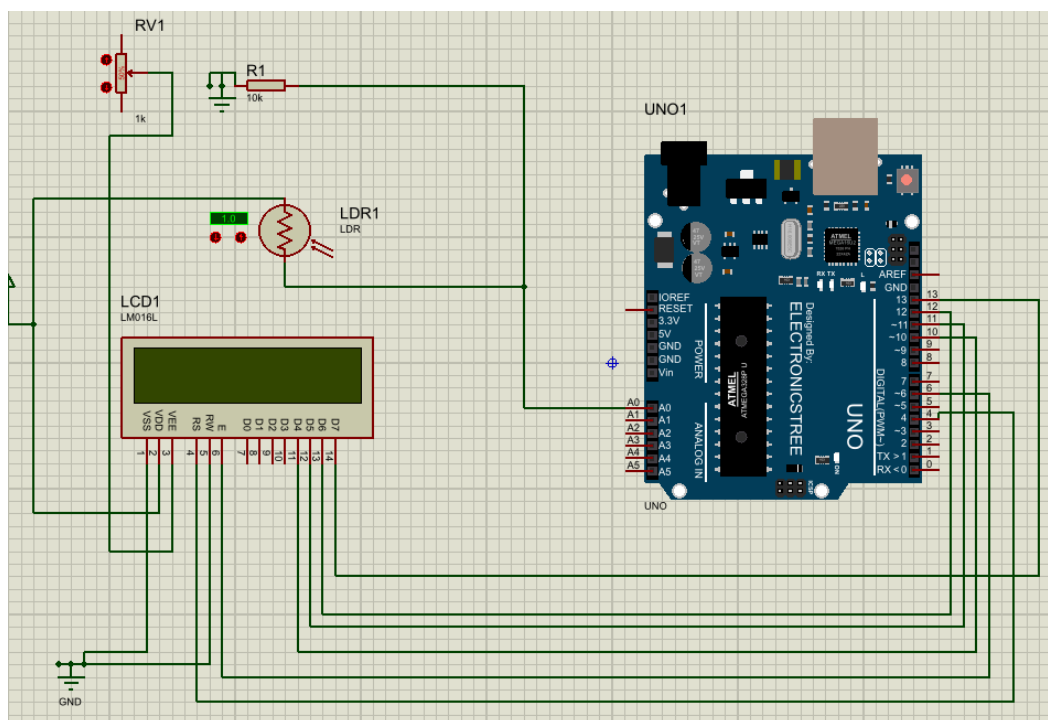
```
sensors_event_t event1, event2;  
accel1.getEvent(&event1);  
accel2.getEvent(&event2);  
  
Serial.print("S1 X: "); Serial.print(event1.acceleration.x);  
Serial.print("  Y: "); Serial.print(event1.acceleration.y);  
Serial.print("  Z: "); Serial.println(event1.acceleration.z);  
  
Serial.print("S2 X: "); Serial.print(event2.acceleration.x);  
Serial.print("  Y: "); Serial.print(event2.acceleration.y);  
Serial.print("  Z: "); Serial.println(event2.acceleration.z);  
  
delay(500);  
}
```

Experiment - 5

Objective:

To measure light intensity using an photoresistor (LDR) and study the variation of resistance with light, and to implement basic analog sensing circuits.

Circuit -



Code -

```
#include <LiquidCrystal.h>

char array1[]="  EPICTAC test  "; //the string to print on the LCD
char array2[]="  DATA: 0000  "; //the string to print on the LCD
int tim = 50;                      //the value of delay time

// initialize the library with the numbers of the interface pins
LiquidCrystal lcd(4, 6, 10, 11, 12, 13);

int photoresistorPin = 0;           // potentiometer wiper (middle terminal) connected
to analog pin 3

void setup()
{
    lcd.begin(16, 2);              // set up the LCD's number of columns and rows:
    lcd.clear();                   //Clears the LCD screen and positions the cursor in the upper-
    left corner
```



```
lcd.setCursor(0,0); // set the cursor to column 15, line 0
for (int positionCounter1 = 0; positionCounter1 < 16; positionCounter1++)
{
    lcd.print(array1[positionCounter1]); // Print a message to the LCD.
    delay(tim);    //wait for 250 microseconds
}
}

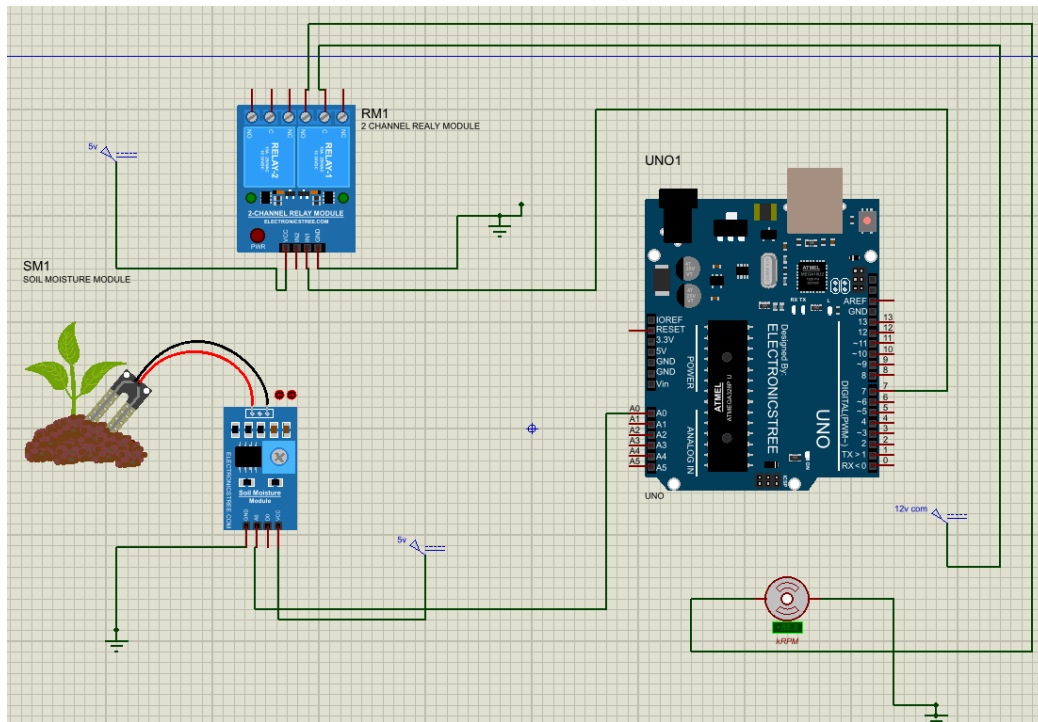
void loop()
{
    array2[9]=analogRead(photoresistorPin)/1000+0x30;    //Take one thousand - bit data
    array2[10]=analogRead(photoresistorPin)/100%10+0x30; //Take one hundred - bit data
    array2[11]=analogRead(photoresistorPin)/10%10+0x30; //Take ten-bit data
    array2[12]=analogRead(photoresistorPin)%10+0x30;    //Take a bit of data
    lcd.setCursor(0,1); // set the cursor to column 15, line 1
    for (int positionCounter3 = 0; positionCounter3 < 16; positionCounter3++)
    {
        lcd.print(array2[positionCounter3]);    // Print a message to the LCD.
        delay(tim);    //wait for 250 microseconds
    }
}
```

Experiment - 6

Objective:

To control electrical devices using a relay module and understand the concept of electrical isolation and switching.

Circuit -



Code -

```
#define sensorPin A0
#define relay1 7    // Relay channel 1
int moistureValue = 0;

void setup()
{
    pinMode(relay1, OUTPUT);
    // Relay OFF initially
    digitalWrite(relay1, HIGH);
    Serial.begin(9600);
}

void loop()
```

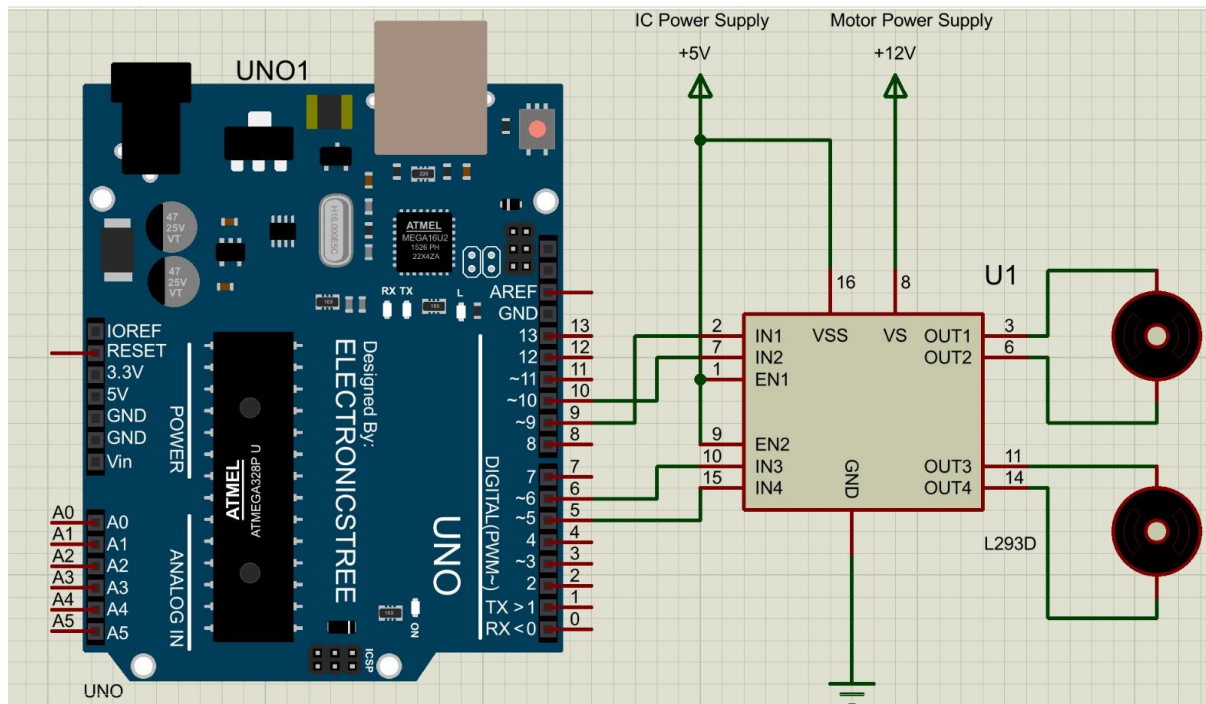
```
{  
  
    // Read soil moisture value  
    moistureValue = analogRead(sensorPin);  
    Serial.print("Moisture Value: ");  
    Serial.println(moistureValue);  
    // Dry soil condition  
    if (moistureValue > 600)  
    {  
        digitalWrite(relay1, LOW);    // Relay ON  
        Serial.println("Pump ON");  
    }  
    else  
    {  
        digitalWrite(relay1, HIGH);    // Relay OFF  
        Serial.println("Pump OFF");  
    }  
    delay(1000);  
}
```

Experiment - 7

Objective:

To simulate and control the direction and speed of a DC motor using an H-Bridge circuit in Proteus configurations.

Circuit -



Code -

```
const int motorPin1 = 5; // Pin 14 of L293
const int motorPin2 = 6; // Pin 10 of L293
```

```
//Motor B
```

```
const int motorPin3 = 10; // Pin 7 of L293
const int motorPin4 = 9; // Pin 2 of L293
```

```
//This will run only one time.
```

```
void setup(){
```

```
//Set pins as outputs
```

```
pinMode(motorPin1, OUTPUT);
```

```
pinMode(motorPin2, OUTPUT);
```

```
pinMode(motorPin3, OUTPUT);
```



```
pinMode(motorPin4, OUTPUT);

//This code will turn Motor A clockwise for 2 sec.
digitalWrite(motorPin1, HIGH);
digitalWrite(motorPin2, LOW);
digitalWrite(motorPin3, LOW);
digitalWrite(motorPin4, LOW);
delay(2000);

//This code will turn Motor A counter-clockwise for 2 sec.
digitalWrite(motorPin1, LOW);
digitalWrite(motorPin2, HIGH);
digitalWrite(motorPin3, LOW);
digitalWrite(motorPin4, LOW);
delay(2000);

//This code will turn Motor B clockwise for 2 sec.
digitalWrite(motorPin1, LOW);
digitalWrite(motorPin2, LOW);
digitalWrite(motorPin3, HIGH);
digitalWrite(motorPin4, LOW);
delay(2000);

//This code will turn Motor B counter-clockwise for 2 sec.
digitalWrite(motorPin1, LOW);
digitalWrite(motorPin2, LOW);
digitalWrite(motorPin3, LOW);
digitalWrite(motorPin4, HIGH);
delay(2000);

//And this code will stop motors
digitalWrite(motorPin1, LOW);
digitalWrite(motorPin2, LOW);
digitalWrite(motorPin3, LOW);
digitalWrite(motorPin4, LOW);
}

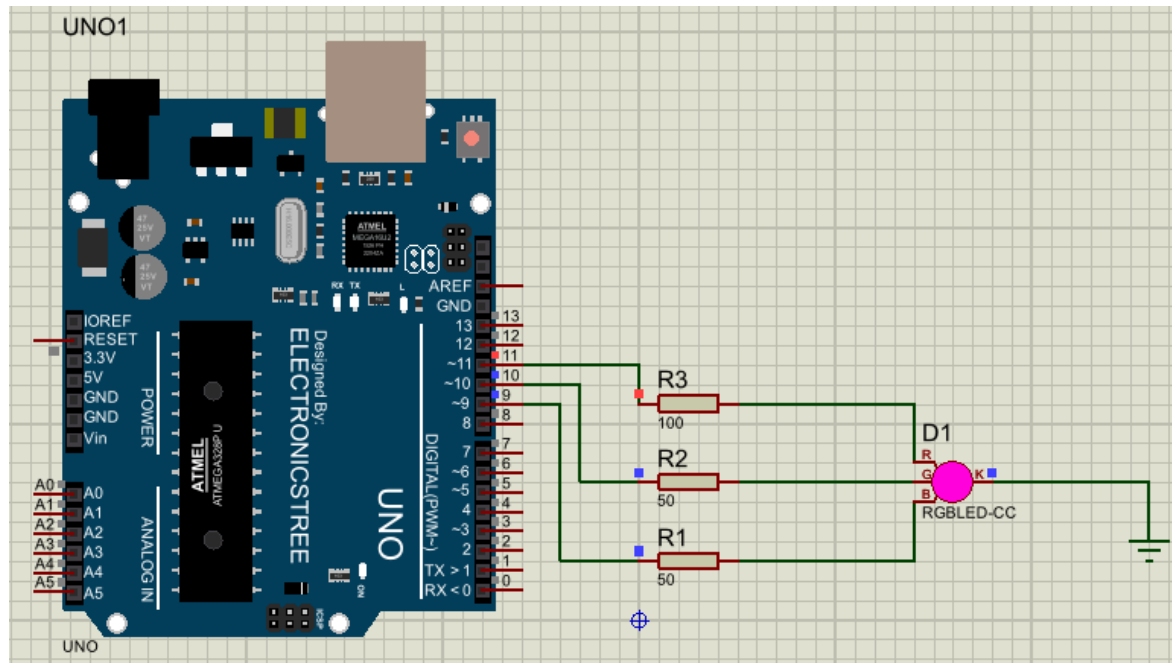
void loop(){
}
```

Experiment - 8

Objective:

To generate different colors by controlling the intensity of Red, Green, and Blue LEDs using PWM techniques.

Circuit -



Code -

```
int redPin= 9;
int greenPin = 10;
int bluePin = 11;

void setup() {
    pinMode(redPin, OUTPUT);
    pinMode(greenPin, OUTPUT);
    pinMode(bluePin, OUTPUT);
}

void loop() {
    setColor(255, 0, 0); // Red Color
    delay(1000);
    setColor(0, 255, 0); // Green Color
    delay(1000);
```




```
setColor(0, 0, 255); // Blue Color
delay(1000);
setColor(255, 255, 255); // White Color
delay(1000);
setColor(170, 0, 255); // Purple Color
delay(1000);
}

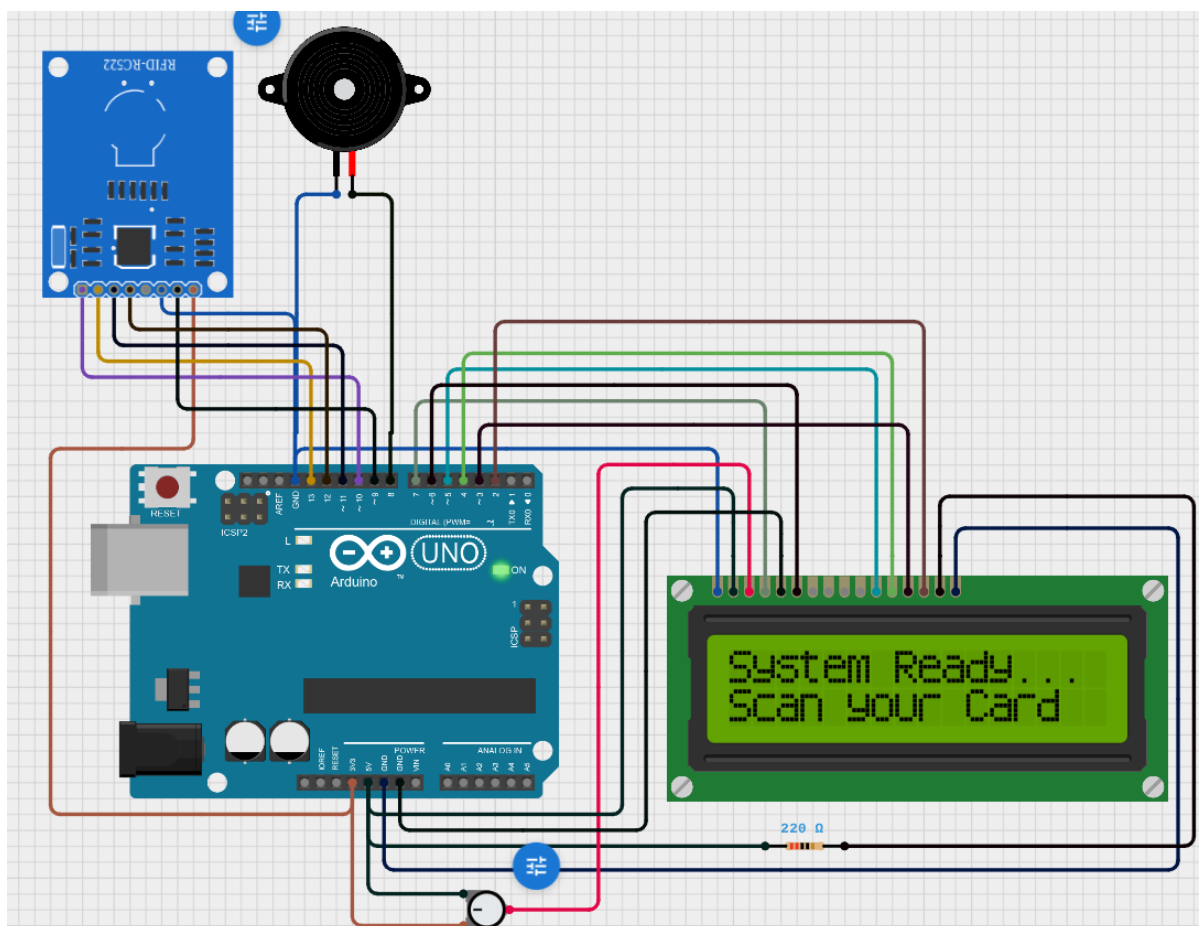
void setColor(int redValue, int greenValue, int blueValue) {
    analogWrite(redPin, redValue);
    analogWrite(greenPin, greenValue);
    analogWrite(bluePin, blueValue);
}
```

Experiment - 9

Objective:

To interface an RFID reader and read data from RFID tags for identification and access control applications.

Circuit -



Code -

```
#include <SPI.h>
#include <MFRC522.h>
#include <LiquidCrystal.h>

// --- Pin Definitions ---
#define RST_PIN          9
#define SS_PIN           10
#define BUZZER_PIN       8

// --- Initialize Components ---
```



```
// RFID Instance
MFRC522 mfrc522(SS_PIN, RST_PIN);

// LCD Instance (RS, E, D4, D5, D6, D7)
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);

void setup() {
    // Initialize Serial Monitor for debugging
    Serial.begin(9600);

    // Initialize SPI bus and RFID module
    SPI.begin();
    mfrc522.PCD_Init();

    // Initialize Buzzer
    pinMode(BUZZER_PIN, OUTPUT);
    digitalWrite(BUZZER_PIN, LOW);

    // Initialize LCD
    lcd.begin(16, 2);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("System Ready...");
    lcd.setCursor(0, 1);
    lcd.print("Scan your Card");

    Serial.println("System Ready. Waiting for card...");
}

void loop() {
    // 1. Look for new cards
    if ( ! mfrc522.PICC_IsNewCardPresent() ) {
        return;
    }

    // 2. Select one of the cards
```

```
if ( ! mfrc522.PICC_ReadCardSerial()) {  
    return;  
}  
  
// 3. Card detected! Beep the buzzer  
digitalWrite(BUZZER_PIN, HIGH);  
delay(100); // Beep duration  
digitalWrite(BUZZER_PIN, LOW);  
  
// 4. Clear LCD and prep for UID  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("Card Detected!");  
lcd.setCursor(0, 1);  
lcd.print("UID: ");  
  
Serial.print("Card UID:");  
  
// 5. Read and display the UID  
String content = "";  
for (byte i = 0; i < mfrc522.uid.size; i++) {  
    // Print to Serial Monitor  
    Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");  
    Serial.print(mfrc522.uid.uidByte[i], HEX);  
  
    // Append to String for LCD  
    content.concat(String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " "));  
    content.concat(String(mfrc522.uid.uidByte[i], HEX));  
}  
  
Serial.println();  
content.toUpperCase();  
  
// Print UID to LCD (removing the leading space)  
lcd.print(content.substring(1));
```



```
// 6. Halt PICC (stop reading the same card repeatedly)
mfrc522.PICC_HaltA();

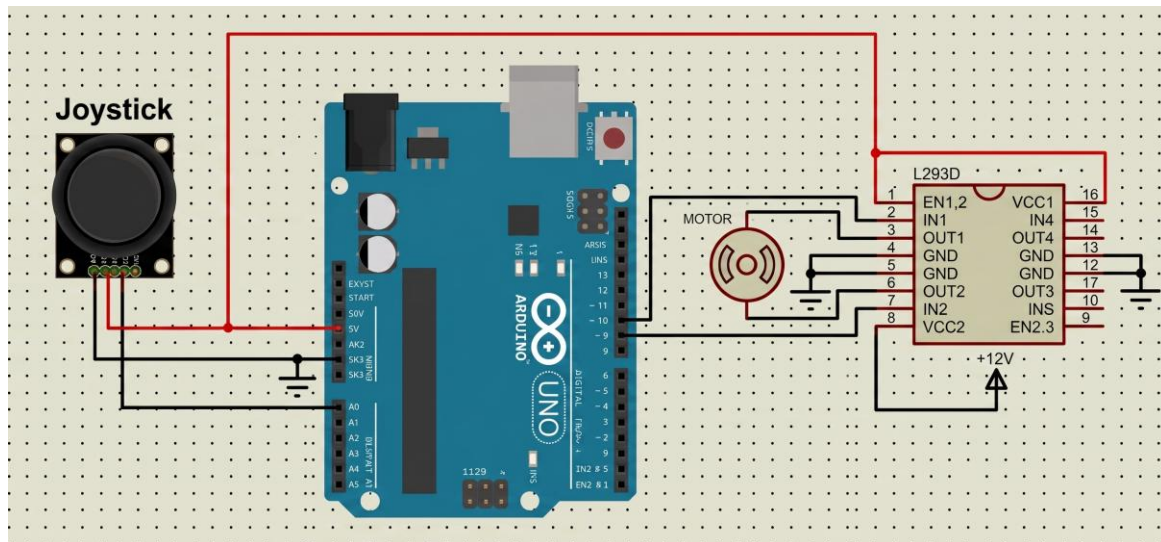
// Wait a few seconds before resetting the screen
delay(3000);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("System Ready...");
lcd.setCursor(0, 1);
lcd.print("Scan your Card");
}
```

Experiment – 10

Objective:

To control of a DC motor using a joystick, and to understand analog input-based motor control.

Circuit –



Code –

```
// Define the pins based on your circuit diagram
const int joyPin = A0;    // Joystick VRx connected to Analog 0
const int motorPin1 = 10; // L293D IN1 connected to PWM pin 10
const int motorPin2 = 9;  // L293D IN2 connected to PWM pin 9

void setup() {
    pinMode(motorPin1, OUTPUT);
    pinMode(motorPin2, OUTPUT);
    // Initialize serial communication for debugging (optional)
    Serial.begin(9600);
}

void loop() {
    // Read the joystick analog value (ranges from 0 to 1023)
    int joyValue = analogRead(joyPin);
    int motorSpeed = 0;
    // Center position is typically around 512.
```

```
// We create a "deadzone" between 470 and 550 so the motor rests completely when untouched.
```

```
if (joyValue >= 470 && joyValue <= 550) {
```

```
    // Stop the motor
```

```
    analogWrite(motorPin1, 0);
```

```
    analogWrite(motorPin2, 0);
```

```
}
```

```
// Joystick pushed in one direction (towards 0)
```

```
else if (joyValue < 470) {
```

```
    // Map the joystick value (470 down to 0) to a PWM speed (0 to 255)
```

```
    motorSpeed = map(joyValue, 470, 0, 0, 255);
```

```
    // Spin motor forward
```

```
    analogWrite(motorPin1, motorSpeed);
```

```
    analogWrite(motorPin2, 0);
```

```
}
```

```
// Joystick pushed in the opposite direction (towards 1023)
```

```
else if (joyValue > 550) {
```

```
    // Map the joystick value (550 up to 1023) to a PWM speed (0 to 255)
```

```
    motorSpeed = map(joyValue, 550, 1023, 0, 255);
```

```
    // Spin motor backward
```

```
    analogWrite(motorPin1, 0);
```

```
    analogWrite(motorPin2, motorSpeed);
```

```
}
```

```
// Optional: Print values to the Serial Monitor to verify joystick readings
```

```
// Serial.print("Joystick Value: ");
```

```
// Serial.print(joyValue);
```

```
// Serial.print(" | Motor Speed: ");
```

```
// Serial.println(motorSpeed);
```

```
// Small delay for stability
```

```
delay(10);
```

```
}
```